UI/UX Design Documentation

Project: Tabular RAG Assistant (Heart Disease Dataset)

Inspiration Reference: "Rotana Signs - Landmark Signage Website 3D UI Design" (Dribbble Shot #27236185)

1. Design Philosophy

The UI will adopt the "Landmark Signage" and "3D Spatial" concepts from the reference. For a medical data assistant, this translates to a highly structured, architectural layout. Instead of flat text, the semantic patient records (retrieved from the CSV) will be rendered as sleek, 3D-extruded glassmorphic cards. The typography will be bold and wayfinding-inspired, guiding the user's eye directly to the most critical health metrics.

2. Color Palette

To merge the bold signage aesthetic with a clinical, trustworthy medical theme:

- **Spatial Background:** `#0A0C10` (Deep Onyx) - Creates an infinite dark canvas to make 3D elements pop.
- **Structural Panels:** `#13161D` (Matte Charcoal) - Used for the chat container and sidebars.
- **Signage Typography (Primary Text):** `#FFFFFF` (Pure White) - High contrast for bold headings.
- **Data Text (Secondary):** `#94A3B8` (Slate Blue/Gray) - For less critical metadata.
- **Primary 3D Accent:** `#00E5FF` (Clinical Cyan) - Used for glowing edges on the 3D data cards and the "Send" button.
- **Medical/Alert Accent:** `#FF3366` (Neon Crimson) - Used sparingly for heart-related iconography and the out-of-context guardrail alerts.

3. Typography

Drawing from the "Signage" inspiration, the typography must be structural and instantly readable.

- **Display/Headers:** *Clash Display* or *Space Grotesk*. (Provides that bold, wide, architectural signage look).
- **Tabular Data & Body:** *Inter* or *Roboto Mono*. (Ensures numbers like `140 BP` or `289 Cholesterol` are perfectly aligned and highly legible).

4. Core 3D UI Components & Layouts

4.1 The Spatial Chat Interface (Center Stage)

- **Input Console:** Anchored at the bottom, styled like a physical control panel rather than a flat input box. It has a slight drop-shadow and a subtle top inner-shadow to look embedded in the screen.
- **Message Bubbles:** * *User:* Flat, `#13161D` blocks with crisp cyan text.
  - *AI (Ollama):* Minimalist text floating directly on the `#0A0C10` background, mimicking large digital signage.

4.2 3D Patient Record Cards (The "Landmarks")

- **Concept:** When Chroma retrieves the top-K relevant rows from the CSV, they do not appear as plain text citations. Instead, they appear in a designated "Data Space" (right sidebar on PC) as 3D-extruded cards.
- **Visuals:** The cards have a glassmorphic face (`backdrop-blur`) with a simulated 3D edge/bevel.
- **Data Visualization:** Inside the 3D card, key CSV metrics (Age, Sex, Cholesterol, MaxHR) are displayed in a clean grid. High-risk metrics (e.g., `HeartDisease: 1`) pulse with a subtle `#FF3366` glow.

4.3 Interactive 3D Canvas (Optional but highly recommended)

- In the background of the app, a slow-rotating, abstract 3D wireframe (e.g., a subtle geometric heart or a data point cloud representing the vector space) reacts to the user's mouse movements, tying into the 3D theme of the Rotana reference.

5. RAG Feature Mapping to UI

RAG Backend Feature	Frontend 3D UI Execution
CSV Ingestion State	A bold, 3D animated "Processing Data..." typography block takes over the screen. A progress bar fills with Clinical Cyan as LangChain converts rows to semantic text.
Semantic Retrieval (Chroma)	As Ollama generates the answer, the 3D Patient Record Cards (e.g., *Patient 44: 40yo Male*) slide in from the right z-axis (scaling up from the background) to act as visual proof.

Out-of-Context Guardrail	**Crucial UI Shift:** If the cosine distance threshold (>0.5) is hit, all 3D elements dim. The UI shifts to a flat, brutalist error state. The text *"Sorry I can't find anything relevant to this question"* appears in bold, strict typography. No data cards are rendered.

6. Universal Responsive Strategy

To translate this 3D signage concept across devices without performance lag:

6.1 PC / Desktop (> 1024px)

- **Layout:** 60/40 Split.
  - *Left 60%:* The spatial chat interface.
  - *Right 40%:* The 3D Data Gallery where retrieved CSV rows stack diagonally in 3D space. Hovering over a 3D card brings it to the front z-index and expands the full metadata.

6.2 Tablet (768px - 1024px)

- **Layout:** Chat takes 100% width. The 3D Patient Record cards move to a horizontal, swipeable carousel docked just above the chat input box.

6.3 Mobile (< 768px)

- **Performance:** True 3D CSS transforms are reduced to 2.5D (subtle drop shadows and overlap) to save battery and rendering overhead.
- **Layout:** Full-screen chat (`100dvh`).
- **Data Cards:** When data is retrieved, a floating action button (FAB) labeled "View Records (3)" appears. Tapping it slides up a structured "Bottom Sheet" where the tabular data is